

AGH UNIVERSITY OF KRAKÓW

Faculty of Electrical Engineering, Automatics, Computer Science and Biomedical Engineering

Bachelor of Engineering thesis

Embedded Implementation and Comparison of Attitude Estimation Methods

Implementacja na urządzenie wbudowane algorytmów odtwarzania orientacji i ich porównanie

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1. Abstract

The main goal of this thesis is to compare various attitude estimation algorithms (with a North unreferenced Yaw) using data from a 6-degree-of-freedom MEMS inertial measurement unit (6DOF IMU). The comparison will be made based on two main criteria, estimation accuracy and computation cost.

The experimental evaluation will include the following algorithms:

– Complementary filter

– Mahony filter

– Extended Kalman Filter (EKF)

– Madgwick filter

The estimation accuracy will be measured under different conditions and at different IMU sampling frequencies. These data will be interpreted using the mentioned algorithms and compared with a reference attitude measurement (obtained by a reference sensor or a motion capture system). The test scenarios will cover measurements in both static conditions (with low external interference) and dynamic conditions (with linear or centrifugal accelerations added). The tests should enable comparison of how different algorithms handle noise directly from sensors and how those filters react to external noise introduced by the environment.

To compare the computational cost, the selected algorithms will be implemented and benchmarked on an STM32 family microcontroller in the C programming language. This will enable a direct measurement of the time or memory overhead introduced by the filter.

1.1. Introduction

2. Methods of Attitude Representation

An attitude of a body is a description of the angular position of its body (senor) frame relative to the world (Earth) reference frame.

The most commonly known method for expressing the attitude of a body is through the use of Euler angles. This method describes the attitude as a sequence of three rotations around three separate axes of reference. Although this method is simple and very intuitive, it is also arguably the most problematic of all the methods mentioned here. The biggest problem of Euler angles representation is the phenomenon of gimbal lock. It appears when two rotation axes become parallel, resulting in the loss of one degree of freedom. This can lead to singularities in the mathematical representation, which in turn causes difficulties and discontinuities in attitude representation. In addition, calculating the rotations of the vectors between two frames described with Euler angles involves the use of computationally expensive trigonometric functions. Additionally, because rotations around each axis are calculated separately, the order of rotations matters and gives a different result based on chosen convention.

Most of the problems of Euler angles can be solved by using rotation matrices. A rotation matrix is a 3x3 orthogonal matrix with it's determinant equal to 1, which represents a rotation in three-dimensional space. This method does not suffer from gimbal lock and provides a unique way of representing rotations, at a cost of reducing intuitiveness of the representation. Rotating a vector using a rotation matrix requires only matrix multiplication. The biggest issue with rotation matrices is their memory footprint. They require 9 parameters to represent a rotation, which is more than necessary and leads to additional numeric complexity and instability. The matrixes will also accumulate numerical errors during rotation concatenation and will require re-orthogonalization to maintain their properties.

Because of that, one more improvement can be made when it comes to attitude representation. It can be achieved through the use of unit quaternions. Quaternions bring the same improvements, over the Euler angles as rotation matrices, while reducing the memory footprint to less than half. They also reduce the number of multiplications required for vector rotation and offer almost as efficient vector rotation as rotation matrices. Quaternions still suffer from the same numerical issues as rotation matrices, but maintaining the unit norm requires much less computational effort. This makes them a natural choice for representing rotations.

Table 2.1. Comparison of attitude representation methods

Method	Advantage	Disadvantage
Euler Angles	Intuitive, low memory footprint	Gimbal lock, computationally expensive
Rotation Matrices	No gimbal lock, unique representation, efficient rotation	Memory footprint, requires re-orthogonalization
Quaternions	No gimbal lock, almost unique representation, efficient concatenation and compared to matrix - reduced memory footprint, and almost as efficient rotation	Requires normalization, sign ambiguity, slightly higher memory footprint than Euler angles

2.1. Representing Rotations as Quaternions

Quaternions are defined as a four-dimensional extension of complex numbers consisting of one real part and three imaginary parts. A quaternion can be represented as:

$$\boldsymbol{q} = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} = [q_0\ q_1\ q_2\ q_3]^T, \quad \boldsymbol{q} \in \mathbb{H} \tag{2.1}$$

where a, b, c, d are real numbers and i, j, k are the basis vectors. The set of all quaternions is denoted by $\mathbb{H}$, for William Rowan Hamilton, who first introduced quaternions in 1843. The neutral element of this set is the unit quaternion:

$$\boldsymbol{q}_I = 1 + 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = [1\ 0\ 0\ 0]^T \tag{2.2}$$

Other common notation for quaternions is to represent them as a scalar part and a vector part:

$$\boldsymbol{q} = [s, \boldsymbol{v}] \in \mathbb{R} \times \mathbb{R}^3, \quad s = q_0, \boldsymbol{v} = [q_1\ q_2\ q_3]^T \tag{2.3}$$

Multiplication between quaternions, called a Hamilton product, is defined as:

$$\boldsymbol{q}_1 \otimes \boldsymbol{q}_2 = \begin{bmatrix} s_1 s_2 - \boldsymbol{v}_1^T \boldsymbol{v}_2 \\ s_1 \boldsymbol{v}_2 + s_2 \boldsymbol{v}_1 + \boldsymbol{v}_1 \times \boldsymbol{v}_2 \end{bmatrix} \tag{2.4}$$

Quaternion conjugate is defined as:

$$\boldsymbol{q}^* = \begin{bmatrix} s \\ -\boldsymbol{v} \end{bmatrix} \tag{2.5}$$

For a quaternion $\boldsymbol{q}$ to represent a valid rotation, it must have a unit norm:

$$\|\boldsymbol{q}\| = \sqrt{\boldsymbol{q}^* \otimes \boldsymbol{q}} = \sqrt{\boldsymbol{q} \otimes \boldsymbol{q}^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} = 1 \tag{2.6}$$

Let a quaternion representing a rotation of frame A with respect to frame B be denoted as ${}^B_A\boldsymbol{q}$, and a vector written in frame B be denoted as ${}^B\mathbf{v}$, with $\mathbf{u}$ being a unit vector. Any rotation θ around a unit vector ${}^B\mathbf{u}$ can be represented as a quaternion:

$${}^B_A\boldsymbol{q} = \begin{pmatrix} \cos\frac{\theta}{2} \\ {}^B\mathbf{u} \sin\frac{\theta}{2} \end{pmatrix} \tag{2.7}$$

Every time an orientation is described using the quaternion without specifying the frames, it will be assumed that the quaternion describes the rotation of the world frame with respect to the body frame.

$$\boldsymbol{q} = {}^{B}_{W}\boldsymbol{q} \tag{2.8}$$

A rotation described by a quaternion is unique, up to its sign. Both $\boldsymbol{q}$ and $-\boldsymbol{q}$ represent the same rotation in space. To inverse the rotation described by a quaternion, it's conjugate can be used:

$${}^{A}_{B}\boldsymbol{q} = {}^{B}_{A}\boldsymbol{q}^{*} \tag{2.9}$$

Quaternion rotations can be easily concatenated together using the Hamilton product:

$${}^{C}_{A}\boldsymbol{q} = {}^{B}_{A}\boldsymbol{q} \otimes {}^{C}_{B}\boldsymbol{q} \tag{2.10}$$

Given two quaternions $\boldsymbol{p}$ and $\boldsymbol{q}$, the cosine of the angle Ω subtended by the arc between them is equal to the dot product of the two quaternions:

$$\cos\Omega = \langle \boldsymbol{p}, \boldsymbol{q} \rangle = s_1 s_2 + \boldsymbol{v}_1^T \boldsymbol{v}_2 \tag{2.11}$$

Rotation quaternions can be interpolated using spherical linear interpolation (SLERP):

$$\text{SLERP}(\boldsymbol{q}, \boldsymbol{p}, \alpha) = \frac{\sin(\alpha\Omega)}{\sin(\Omega)}\boldsymbol{p} + \frac{\sin([1-\alpha]\Omega)}{\sin(\Omega)}\boldsymbol{q}, \alpha \in [0, 1] \tag{2.12}$$

This interpolation can also be approximated with a simple linear interpolation (LERP), reducing the computational cost at the cost of the accuracy and loss of the unit norm.

$$\text{LERP}(\boldsymbol{q}, \boldsymbol{p}, \alpha) = (1-\alpha)\boldsymbol{q} + \alpha\boldsymbol{p}, \alpha \in [0, 1] \tag{2.13}$$

To perform any mathematical operation between a vector and a quaternion, it is common practice to first convert the vector into a pure quaternion $\boldsymbol{p}$, i.e., a quaternion with a zero scalar part. The transformed vector can be later extracted from the pure quaternion by taking its vector part.

$${}^{B}\boldsymbol{p} = \begin{bmatrix} 0 \\ {}^{B}\boldsymbol{v} \end{bmatrix} \tag{2.14}$$

To perform a rotation of a vector ${}^{B}\boldsymbol{v}$ from frame B to frame A, the following operation can be performed:

$${}^{A}\boldsymbol{p} = {}^{B}_{A}\boldsymbol{q} \otimes {}^{B}\boldsymbol{p} \otimes {}^{B}_{A}\boldsymbol{q}^{*} \tag{2.15}$$

The quaternion can also be converted into a rotation matrix:

$$\boldsymbol{R}(\boldsymbol{q}) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \tag{2.16}$$

Or alternatively, after enforcing the unit norm of the quaternion, as:

$$\boldsymbol{R}(\boldsymbol{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix} \tag{2.17}$$

2.2. Chosen frame of reference

The global frame of reference will be represented using NED (North-East-Down) coordinate system. The NED system is a right-handed local tangent plane coordinate system. The x-axis of this system points towards the north, the y-axis points towards the east, and the z-axis points downward, aligning with the local gravity vector.

Because of common quaternion conventions, every quaternion representing a rotation will describe an orientation of the world frame relative to the body frame. The body frame will be denoted as $\{B\}$, the world frame as $\{W\}$. Additionally, any estimator will provide only an approximation of the real world frame rotation, this can be fought of as a separate estimator frame $\{E\}$.

3. Attitude from IMU Measurements

A typical MEMS IMU consists of 2 separate sensors. A rate gyroscope, measuring angular velocity along the 3 sensor frame axis, in rad/s. As well as a 3-axis accelerometer, measuring linear acceleration along the same three axes, in m/s^2. Neither of those two sensors can be used alone to provide a reliable attitude observation in real conditions.

3.1. Attitude from Accelerometer

The measurement model for accelerometer observations a_y can be represented as:

$$\boldsymbol{a}_y = {}^{W}_{B}\boldsymbol{R}(\dot{\boldsymbol{v}} + \boldsymbol{g}) + \boldsymbol{b}_a + \boldsymbol{\mu}_a \tag{3.1}$$

Where:

- ${}^{W}_{B}\boldsymbol{R}$ – Rotations matrix representing transformation from world to body frame
- $\boldsymbol{g}$ – Gravity vector in word frame
- $\dot{\boldsymbol{v}}$ – Body frame acceleration
- $\boldsymbol{b}_a$ – Bias of the accelerometer
- $\boldsymbol{\mu}_a$ – Measurement noise, including all unmodeled dynamics of the system

Assuming perfectly static conditions and 0 bias term, the accelerometer measures only the gravitational acceleration vector in the sensor frame. This measurement can be used to calculate a transformation aligning the body frame to the NED frame down direction. It is important to note that this method does not provide full orientation of the body, only the attitude can be estimated, while the heading (yaw) remains unobservable.

In most common cases, the sensor will be subject to additional and often unpredictable external forces, which will make accelerometer only attitude estimation unreliable or even dangerous when used for control purposes.

3.2. Attitude from Gyroscope

Gyroscope measurements Ω_y can be represented as:

$$\begin{aligned} \boldsymbol{\omega}_y &= {}^{B}\boldsymbol{\omega} + \boldsymbol{b}_g + \boldsymbol{\mu}_g \\ \boldsymbol{\omega}_y &= [\omega_x, \omega_y, \omega_z]^T \end{aligned} \tag{3.2}$$

Where:

$^{B}\boldsymbol{\omega}$ – Real angular velocity vector in the sensor frame

$\boldsymbol{b}_g$ – Gyroscope bias, slowly varying over time

$\boldsymbol{\mu}_g$ – Measurement noise, including all unmodeled dynamics of the system

Assuming an initial orientation, which cannot be measured by the gyroscope, the orientation can be estimated by integrating the angular velocity over time. This method gives a full orientation estimate, which is mostly unaffected by external linear accelerations. However, due to measurement noise, numerical integration errors and, most importantly, the slowly varying bias, this estimate will drift over time, making it unreliable for long term use.

A commonly used quaternion derivative model, that can be used to integrate the gyroscope measurements, is given by:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \otimes \boldsymbol{p}(^{B}\boldsymbol{\Omega}) \tag{3.3}$$

3.3. Attitude from Sensor Fusion

Although achieving a reliable and universal attitude estimation from either of the two sensors alone is not possible, combining the measurements from both sensors can provide a much more accurate and stable estimate. The gyroscope measurement is much more reliable and accurate in short term observations, so its going to serve as the ”base” of the algorithm. The accelerometer measurement will correct the errors that accumulate in the gyroscope based estimate, as well as providing an initial orientation estimate.

4. Chosen Attitude Estimation Algorithms

4.1. Mahony filter

The Mahony filter is a name commonly used to refer to a group of complementary filters derived by Robert Mahony in [1]. The main idea behind those algorithms is to provide a computationally inexpensive solution for estimating attitude from inexpensive MEMS IMU sensors. Mahony *et. al.* expands on the idea of the classic complementary filter by introducing a feedback term to the gyroscope measurement, which is an integral of the error between the accelerometer measurement and the expected gravity vector in the sensor frame. This feedback term helps to correct the drift in the gyroscope based estimate, while still maintaining the computational simplicity of a complementary filter. In this work, the *explicit complementary filter* version of the algorithm proposed in [1] will be used.

To derive the correction term for the filter, Madgwick uses an error that is the relative rotation from the body frame to the estimator frame. In practice, the real body frame rotation is unknown, so instead an estimated rotation derived from available measurements R_y is used.

$$E_{tr} = \frac{1}{2}\text{tr}(\boldsymbol{I}_3 - \hat{\boldsymbol{R}}^T \boldsymbol{R_y}) = 1 - \cos(\theta) \tag{4.1}$$

Where θ is the angle associated with the rotation between those frames. The filter dynamics, in quaternion form, is given by:

$$\begin{aligned} \dot{\hat{\boldsymbol{q}}} &= \frac{1}{2}\hat{\boldsymbol{q}} \otimes \boldsymbol{p}(\boldsymbol{\omega}_y - \hat{\boldsymbol{b}} + k_p\boldsymbol{\omega}) \\ \dot{\hat{\boldsymbol{b}}} &= -k_I\boldsymbol{\omega} \end{aligned} \tag{4.2}$$

The goal of the estimator will be to find a term $\boldsymbol{\omega}$ that will drive this error to 0. The correction term introduced by Mahony *et. al.* is:

$$\boldsymbol{\omega} = \text{vex}(\mathbb{P}_a(\hat{\mathbf{R}}^T\mathbf{R}_y)) = sin(\theta)\mathbf{u} \tag{4.3}$$

Where:

$$\boldsymbol{\omega}_\times = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}, \boldsymbol{\omega} \in \mathbb{R}^3 \tag{4.4}$$

$$vex(\boldsymbol{\omega}_\times) = \boldsymbol{\omega} \tag{4.5}$$

$$\mathbb{P}_a(\boldsymbol{A}) = \frac{1}{2}(\boldsymbol{A} - \boldsymbol{A}^T) \tag{4.6}$$

With normalized accelerometer measurement v and expected gravity vector $\hat{v} = \mathbf{R}(\hat{\mathbf{q}})^T\mathbf{g}$, $g = [0, 0, 1]^T$, the correction term can be expressed as:

$$\boldsymbol{\omega} = v \times \hat{v} \tag{4.7}$$

With the correction term defined, the full explicit complementary filter dynamics can be implemented as follows:

$$\begin{aligned} \dot{\hat{\boldsymbol{q}}} &= \frac{1}{2}\hat{\boldsymbol{q}} \otimes \boldsymbol{p}(\boldsymbol{\omega}_y - \hat{\boldsymbol{b}} + k_p\boldsymbol{\omega}) \\ \dot{\hat{\boldsymbol{b}}} &= -k_I\boldsymbol{\omega} \\ \boldsymbol{\omega} &= k_I(\boldsymbol{v} \times \hat{\boldsymbol{v}}) \end{aligned} \tag{4.8}$$

The estimator can be parameterized by two gains, k_p and k_I. The parameter k_p controls the instantaneous filter correction. If its too high, the filter will be very sensitive to accelerometer noise, while if its too low, the filter will be slow to correct for gyroscope drift, or even diverge all together. The parameter k_I controls the integral correction term, which is responsible for estimating the gyroscope bias.

4.2. AQUA Complementary filter

In [2] Roberto G. Valenti *et. al.* describe an orientation estimation algorithm, which is an interesting variation on a standard complementary filter algorithm. The main idea behind his approach is to provide an estimation as the algebraic solution of a system from inertial and optionally magnetic observations. During the correction step, the algorithm finds the tilt quaternion from the accelerometer and the heading quaternion from the magnetometer observations separately. This allows to avoid the problem of corrections originating from the magnetometer measurements affecting the attitude estimation.

The dynamics of the filter prediction step, as proposed in [2], but adjusted for notation conventions, is given by:

$$\dot{{}^B_E\hat{\boldsymbol{q}}}_{\omega,t_k} = \frac{1}{2}\hat{\boldsymbol{q}}_{\omega,t_{k-1}} \otimes \boldsymbol{p}(\boldsymbol{\Omega}_{y,t_k}) \tag{4.9}$$

This equation can be computed numerically, using a simple Euler integration method:

$${}^B_E\hat{\boldsymbol{q}}_{\omega,t_k} = \hat{\boldsymbol{q}}_{\omega,t_{k-1}} + \dot{\hat{\boldsymbol{q}}}_{\omega,t_k}\Delta t \tag{4.10}$$

In the correction step, a delta quaternion is computed from the accelerometer observation, which is then used to correct the predicted orientation.

$$\hat{\boldsymbol{q}}_{t_k} = {}^E_W\widehat{\Delta\boldsymbol{q}}_{acc} \otimes {}^B_E\hat{\boldsymbol{q}}_{\omega,t_k} \tag{4.11}$$

The delta quaternion $\Delta\boldsymbol{q}_{acc}$satisfies the following system of equations:

$$\begin{cases} 2(\Delta q_{1,acc}\Delta q_{3,acc} + \Delta q_{0,acc}\Delta q_{2,acc}) & = G_x \\ 2(\Delta q_{2,acc}\Delta q_{3,acc} + \Delta q_{0,acc}\Delta q_{1,acc}) & = G_y \\ \Delta q^2_{0,acc} - \Delta q^2_{1,acc} - \Delta q^2_{2,acc} + \Delta q^2_{3,acc} & = G_z \end{cases} \tag{4.12}$$

Where $\boldsymbol{G} = \boldsymbol{R}({}^B_W\hat{\boldsymbol{q}})\boldsymbol{g} = [G_x, G_y, G_z]^T$, $g = [0, 0, 1]^T$, is the gravity vector predicted from the orientation estimate. After defining the component $\Delta q_{3,acc} = 0$ or $\Delta q_{2,acc} = 0$, the solution to this system of can

be found, one for each case. Two solutions were given to avoid singularities in the solution, depending on the value of G_z:

$$\Delta \boldsymbol{q}_{acc} = \begin{cases} [\sqrt{\frac{G_z+1}{2}} \quad \frac{G_y}{2G_z+1} \quad -\frac{G_x}{2G_z+1} \quad 0] & , G_z >= 0 \\ [-\frac{G_y}{2(1-G_z)} \quad -\sqrt{\frac{1-G_z}{2}} \quad 0 \quad -\frac{G_x}{2(1-G_z)}] & , G_z < 0 \end{cases} \tag{4.13}$$

Applying the full correction term will instantaneously rotate the predicted orientation to match the tilt observed by the accelerometer. This is undesirable, as the accelerometer measures both the gravity vector and the linear acceleration of the sensor. To avoid this, $\Delta \boldsymbol{q}_{acc}$ is interpolated with the identity quaternion, using a gain parameter k_{acc}. Valenti *et. al.* propose using LERP interpolation for relatively small corrections. The metric given for this selection is the cosine of the subtended angle Ω between the interpolated quaternions. The suggested threshold over which *LERP* should be used is $cos(\Omega) > 0.9$, *SLERP* should be used otherwise. When interpolating with a quaternion, $cos(\Omega)$ is equal to the scalar part of the $\Delta \boldsymbol{q}_{acc}$ quaternion. The final correction term becomes:

$$\overset{E}{_W}\widehat{\Delta \boldsymbol{q}}_{acc} = \begin{cases} \frac{\text{LERP}(\boldsymbol{q}_I, \Delta \boldsymbol{q}_{acc}, k_{acc})}{||\text{LERP}(\boldsymbol{q}_I, \Delta \boldsymbol{q}_{acc}, k_{acc})||} & , \Delta q_{0,acc} > 0.9 \\ \text{SLERP}(\boldsymbol{q}_I, \Delta \boldsymbol{q}_{acc}, k_{acc}) & , \Delta q_{0,acc} \leq 0.9 \end{cases} \tag{4.14}$$

4.3. Madgwick filter

In [3] Sebastian O.H. Madgwick presents an estimation algorithm that departs from the traditional observer design, and threats the problem of attitude estimation, directly as an optimization problem. The algorithm uses the accelerometer and magnetometer measurements in an analytically derived gradient descent algorithm, minimizing the gyroscope measurement errors.

The optimization problem from IMU observations is defined as follows:

$$\hat{\boldsymbol{q}}_{k+1} = \hat{\boldsymbol{q}}_k - \mu \frac{\nabla f(\hat{\boldsymbol{q}}_k, \boldsymbol{a}_y)}{||\nabla f(\hat{\boldsymbol{q}}_k, \boldsymbol{a}_y)||} \tag{4.15}$$

$$\nabla f(\hat{\boldsymbol{q}}_k, \boldsymbol{a}_y) = \boldsymbol{J}^T(\boldsymbol{q}_k) f(\hat{\boldsymbol{q}}_k, \boldsymbol{a}_y) \tag{4.16}$$

$$f(\boldsymbol{q}, {}^B\boldsymbol{a}) = \begin{bmatrix} 2(q_1 q_3 - q_0 q_2) - a_x \\ 2(q_0 q_1 + q_2 q_3) - a_y \\ 2(\frac{1}{2} - q_1^2 - q_2^2) - a_z \end{bmatrix} \tag{4.17}$$

$$\boldsymbol{J}(\boldsymbol{q}) = \begin{bmatrix} -2q_2 & 2q_3 & -2q_0 & 2q_1 \\ 2q_1 & 2q_0 & 2q_3 & 2q_2 \\ 0 & -4q_1 & -4q_2 & 0 \end{bmatrix} \tag{4.18}$$

Using this optimization problem, the filter dynamics can be defined as:

$$\begin{aligned} \hat{\boldsymbol{q}}_{est,t} &= \hat{\boldsymbol{q}}_{est,t-1} + \dot{\boldsymbol{q}}_{est,t} \Delta t \\ \dot{\boldsymbol{q}}_{est,t} &= \dot{\boldsymbol{q}}_{\omega,t} - \beta \dot{\hat{\boldsymbol{q}}}_{\epsilon,t} \\ \dot{\hat{\boldsymbol{q}}}_{\epsilon,t} &= \frac{\nabla f(\hat{\boldsymbol{q}}_{est,t-1}, \boldsymbol{a}_y)}{||\nabla f(\hat{\boldsymbol{q}}_{est,t-1}, \boldsymbol{a}_y)||} \end{aligned} \tag{4.19}$$

4.4. Extended Kalman Filter

The Extended Kalman Filter is a generalization of the classic linear Kalman observer [4], that allows it to be applied to nonlinear systems. It is achieved by linearizing the system dynamics and measurement equations around the current state estimate, and then applying the standard Kalman filter equations to the linearized system.

The discrete observer state transition and observation models are defined as follows:

$$\begin{aligned} \boldsymbol{x}_k &= f(\boldsymbol{x}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{w}_{k-1} \\ \boldsymbol{z}_k &= h(\boldsymbol{x}_k) + \boldsymbol{v}_k \end{aligned} \tag{4.20}$$

EKF is a recursive algorithm, which consists of two steps: prediction and correction (or innovation). In the prediction step, the state estimate and error covariance are propagated forward in time using the system dynamics. In the correction step, the state estimate and its error covariance are corrected using the new measurement and calculated Kalman gain.

Prediction:

$$\begin{aligned} \hat{\boldsymbol{x}}_{k|k-1} &= f(\hat{\boldsymbol{x}}_{k-1|k-1}, \boldsymbol{u}_{k-1}) \\ \boldsymbol{P}_{k|k-1} &= \boldsymbol{F}_k \boldsymbol{P}_{k-1|k-1} \boldsymbol{F}_k^T + \boldsymbol{Q}_{k-1} \\ \boldsymbol{F}_k &= \frac{\partial f}{\partial \boldsymbol{x}} |_{\hat{\boldsymbol{x}}_{k-1|k-1}, \boldsymbol{u}_{k-1}} \end{aligned} \tag{4.21}$$

Where:

- $\boldsymbol{P}_{k|k-1}$ – Predicted covariance of the state estimate
- $\boldsymbol{F}_k$ – State transition Jacobian
- $\boldsymbol{Q}_{k-1}$ – Process noise covariance matrix

Innovation:

$$\begin{aligned} \tilde{\boldsymbol{y}}_k &= \boldsymbol{z}_k - h(\hat{\boldsymbol{x}}_{k|k-1}) \\ \boldsymbol{S}_k &= \boldsymbol{H}_k \boldsymbol{P}_{k|k-1} \boldsymbol{H}_k^T + \boldsymbol{R}_k \\ \boldsymbol{K}_k &= \boldsymbol{P}_{k|k-1} \boldsymbol{H}_k^T \boldsymbol{S}_k^{-1} \\ \hat{\boldsymbol{x}}_{k|k} &= \hat{\boldsymbol{x}}_{k|k-1} + \boldsymbol{K}_k \tilde{\boldsymbol{y}}_k \\ \boldsymbol{P}_{k|k} &= (\mathbf{I} - \boldsymbol{K}_k \boldsymbol{H}_k) \boldsymbol{P}_{k|k-1} \\ \boldsymbol{H}_k &= \frac{\partial h}{\partial \boldsymbol{x}} |_{\hat{\boldsymbol{x}}_{k|k-1}} \end{aligned} \tag{4.22}$$

Where:

- $\boldsymbol{y}_k$ – Measurement innovation
- $\boldsymbol{S}_k$ – Innovation covariance
- $\boldsymbol{H}_k$ – Observation model Jacobian
- $\boldsymbol{R}_k$ – Measurement noise covariance matrix
- $\boldsymbol{K}_k$ – Near-optimal Kalman gain

When applying the EKF to the problem of attitude estimation, the gyroscope measurements are used as the model input, while the accelerometer and optionally magnetometer measurements are used for innovation. The state vector $\boldsymbol{x}$ and the input vector $\boldsymbol{u}$ can be defined as follows:

$$\boldsymbol{x} = \boldsymbol{q} = [q_0 \; q_1 \; q_2 \; q_3]^T \tag{4.23}$$

$$\boldsymbol{u} = \boldsymbol{\Omega}_y = [\omega_x, \omega_y, \omega_z]^T \tag{4.24}$$

The state transition function and its Jacobian can be defined using (3.3)

$$f(\boldsymbol{x}_k, \boldsymbol{u}_k) = \boldsymbol{q}_k + \frac{1}{2}\boldsymbol{q}_k \otimes \boldsymbol{p}(\boldsymbol{\Omega}_{y,k})\Delta t \tag{4.25}$$

$$\boldsymbol{F}(\boldsymbol{x}_k, \boldsymbol{u}_k) = \begin{bmatrix} 1 & -\frac{1}{2}\omega_{x,k}\Delta t & -\frac{1}{2}\omega_{y,k}\Delta t & -\frac{1}{2}\omega_{z,k}\Delta t \\ \frac{1}{2}\omega_{x,k}\Delta t & 1 & \frac{1}{2}\omega_{z,k}\Delta t & -\frac{1}{2}\omega_{y,k}\Delta t \\ \frac{1}{2}\omega_{y,k}\Delta t & -\frac{1}{2}\omega_{z,k}\Delta t & 1 & \frac{1}{2}\omega_{x,k}\Delta t \\ \frac{1}{2}\omega_{z,k}\Delta t & \frac{1}{2}\omega_{y,k}\Delta t & -\frac{1}{2}\omega_{x,k}\Delta t & 1 \end{bmatrix} \tag{4.26}$$

With IMU only attitude estimation, the observation $\boldsymbol{z}_k$ represent the accelerometer measurements in the body frame. The measurement estimate can be computed by rotating the global gravity vector ${}^W\boldsymbol{g} = [0 \; 0 \; 1]^T$ into the body frame using the current orientation estimate $\hat{\boldsymbol{q}}_{k|k-1}$:

$$\begin{aligned} \tilde{y}_k &= \boldsymbol{a}_y - h(\boldsymbol{q}_k) \\ h(\boldsymbol{q}_k) &= \mathbf{R}^T(\hat{\boldsymbol{q}}_k)\,{}^W\boldsymbol{g} \end{aligned} \tag{4.27}$$

The Jacobian of the observation model is:

$$\boldsymbol{H}(\boldsymbol{x}_k) = \begin{bmatrix} -2q_2 & 2q_3 & -2q_0 & 2q_1 \\ 2q_1 & 2q_0 & 2q_3 & 2q_2 \\ 2q_0 & -2q_1 & -2q_2 & 2q_3 \end{bmatrix} \tag{4.28}$$

5. Implementation of Selected Algorithms

6. Experimentation

6.1. Equipment

6.2. Methodology

6.3. Results

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